

INTRODUCTION TO PROOFS

DIRECT PROOF

In a direct proof, we assume that P is true and use **axioms**, **definitions**, and previously **proven theorems**, together with **rules of inference** to show that Q must be true.

Straightforward: Leading from hypothesis to the Conclusion

Example #1

Prove that the theorem "If n is odd, then n^2 is odd" by giving direct Proof.

Sol:

$P(n)$: n is odd integer

$Q(n)$: n^2 is odd

Here we assume, $P(n)$ is true and to show that $Q(n)$ is also true.

Assumption:

$P(n)$ is true i.e.,

n is odd integer

It can be written as :

$$n = 2K + 1 \quad \text{for some integer } K \quad (\text{By Definition of odd integer})$$

Now we have to show that

n^2 is odd

As

$$n = 2K + 1$$

$$n^2 = (2K + 1)^2 \quad : \text{squaring both sides}$$

$$n^2 = 4K^2 + 4K + 1$$

$$n^2 = 2(2K^2 + 2K) + 1$$

$$= 2m + 1 \quad \text{where } m = 2K^2 + 2K \text{ is some integer}$$

= odd form of integer

So, n^2 is odd.

and we have proved that the theorem

If n is odd integer, then n^2 is odd

Example #2

Give a direct proof that

"If m and n are both perfect squares, then mn is also perfect square."

Sol :-

Let m and n are perfect squares, so by definition of perfect square m and n can be written as :

$$m = s^2 \quad \text{where } s \text{ is an integer}$$

$$n = t^2 \quad \text{where } t \text{ is an integer}$$

we have to show that

$$mn = (\text{some int.})^2$$

$$\begin{aligned} mn &= s^2 t^2 \\ &= (st)^2 \quad \text{where } st \text{ is an integer} \end{aligned}$$

So it is proved that

If m, n are perfect squares, then mn is also perfect square.

PROOF BY CONTRAPOSITION :

- * **Direct proofs** Lead from hypothesis to the Conclusion
(However, Sometimes, we will reach at dead end)
- * **Proof By Contraposition (or Indirect Proof)** Lead from negation of Conclusion to the negation of hypothesis of the original Conditional Statement.

For statement

$$P \rightarrow Q$$

Contrapositive form

$$\neg Q \rightarrow \neg P$$

We take $\neg Q$ as the hypothesis and using **axioms**, **definitions**, and previous **proven theorems**, together with **rules of inference**, we show that $\neg P$ must follow (i.e. $\neg P$ true).

NOTE : (Rule of Thumb):

First,

Evaluate whether a direct proof looks promising. (If yes)

Begin by expanding the definitions in the **hypothesis**. Start to reason using these hypothesis, together with axioms, and available theorems.

If direct proof does not seem to go anywhere (reach at dead end),

TRY:

the same thing with the proof by **Contraposition**.

i.e.,

assume the Conclusion of the Conditional Statement is **false** and use a **direct proof** to show this implies that the hypothesis of the Conditional Statement must be false

original statement

$$P \rightarrow Q$$

Contrapositive of original

$$\neg Q \rightarrow \neg P$$

Both are Same.

Example #1 (Proof By Contradiction)

Prove that

If n is an integer and $3n+2$ is odd, then n is odd.

Sol:

Direct Proof:

$P(n)$: $3n+2$ is odd

$Q(n)$: n is odd

Assume $P(n)$ is true, and we have to show that $Q(n)$ is also true.

$3n+2$ is odd

then it can be written as:

$$3n+2 = 2K+1 \quad \text{--- (A) where } K \text{ is some integer}$$

and we have to show that $Q(n)$: n is odd.

Can we use (A) to show that n is odd?

But there ~~does~~ ^{does} not seem to be any direct way to conclude from (A), ^{that} n is odd.

Indirect Proof (Proof By Contradiction):

$P(n)$: $3n+2$ is odd

$Q(n)$: n is odd

$$\neg Q \rightarrow \neg P$$

Assumption:

$\neg Q$ is true i.e. n is even

To prove:

$\neg P$ is true i.e. $3n+2$ is even

As n is even

$$\Rightarrow n = 2K \quad \text{where } K \text{ is some integer}$$

and

$$3n+2 = 3(2K)+2$$

$$= 6K+2$$

$$= 2(3K+1)$$

$$= 2m$$

$$= \text{Even}$$

multiple of 2.

where m is integer

As $P \rightarrow Q$ and $\neg Q \rightarrow \neg P$ are equivalent, so we have proved that If $3n+2$ is odd, then n is odd.

So, $3n+2$ is even $\Rightarrow \neg Q \rightarrow \neg P$ is true ^{false.}
Conclusion implies that hypothesis

Example #2 (Proof By Contraposition): Indirect Proof

Prove that If $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Sol:-

There is no obvious way of showing that $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ directly from $n = ab$. So we use Indirect method.

$$P(n): (n = ab)$$

$$Q(n): (a \leq \sqrt{n} \text{ or } b \leq \sqrt{n})$$

$$P \rightarrow Q : \text{DIRECT}$$

$$\neg Q \rightarrow \neg P : \text{INDIRECT}$$

Assumption:

$$\neg Q \text{ is true i.e. } \neg(a \leq \sqrt{n} \text{ or } b \leq \sqrt{n}) \text{ is true}$$

$$= \neg(a \leq \sqrt{n} \text{ or } b \leq \sqrt{n})$$

$$= \neg(a \leq \sqrt{n}) \wedge \neg(b \leq \sqrt{n}) \quad \text{By Demorgan's Law}$$

$$= a > \sqrt{n} \wedge b > \sqrt{n} \text{ is true}$$

and we have to show that

$$\neg P \text{ is true i.e. } \neg(n = ab) \text{ or } n \neq ab$$

$$\text{As } a > \sqrt{n} \text{ and } b > \sqrt{n}$$

$$ab > \sqrt{n} \cdot \sqrt{n} \quad \left\{ \begin{array}{l} \text{multiplying inequalities together} \\ \text{(using the fact that } t > s \text{ if } t > 0, v > 0) \end{array} \right.$$

$$ab > n$$

$$\text{i.e. } ab \neq n \quad (\neg P \text{ is true})$$

~~Because~~
So $\neg Q \rightarrow \neg P$ is true

Negation of the Conclusion of the original Conditional Statement implies that hypothesis of original Conditional Statement is false.

We have proved that

If $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ is true.

$P \rightarrow Q$ and $\neg Q \rightarrow \neg P$ are equivalent

Example #3 (Proof By Contraposition) : Indirect Method

If n is an integer and n^2 is odd, then n is odd.

Sol:

First Attempt : Direct Method.

Suppose P is true i.e

n^2 is odd true

and we have to show $Q(n)$ i.e n is odd true.

Assumption: n^2 is odd

i.e $n^2 = 2K + 1$ for some K

and $n = \pm\sqrt{2K+1}$

using this, ~~and~~ we have to show that n is odd

There seem no obvious approach to show that n is odd from $n = \pm\sqrt{2K+1}$
we use indirect method to prove it.

Indirect Proof : Proof By Contraposition ($\neg Q \rightarrow \neg P$)

$P(n)$: n^2 is odd

$Q(n)$: n is odd

Assumption:

$\neg Q$ is true and then to show that $\neg P$ is true (using direct proof).

$\neg Q$ is true means n is not odd : True

i.e n is even

$\Rightarrow n = 2K$ and to show that $\neg P$: n^2 is even is true.

Squaring both sides

$$n^2 = (2K)^2$$

$$= 4K^2$$

$$= 2(2K^2) \quad m = 2K^2 \text{ is some integer}$$

$$= 2m$$

= even form, so $\neg P$ is true i.e hypothesis is false

So, we have proved that

If n^2 is odd, then n is odd for integer n .

VACUOUS AND TRIVIAL PROOFS

VACUOUS PROOFS:

Vacuous Proofs are often used to establish special cases of Theorems that state that a Conditional statement is true for all positive integers [i.e., a theorem of the Kind $\forall n P(n)$, where $P(n)$ is propositional function]

We can quickly prove that a Conditional statement $P \rightarrow Q$ is true when we know that P is false, because $P \rightarrow Q$ must be true when P is false (no matter what the values of Q).

Example: (Vacuous Proof): hypothesis false case.

Show that the proposition $P(0)$ is true, where

$P(n)$: If $n > 1$, then $n^2 > n$

domain: all integers

Sol:

We can show $P(0)$ is true by vacuous proof.

$P(n)$: If $n > 1$, then $n^2 > n$

$n=0$ $P(0)$: If $0 > 1$, then $0^2 > 0$

Here hypothesis ($0 > 1$) is false, so $P(0)$ must be true, no matter what the values of Conclusion.

VACUOUS PROOF

A proof of $P \rightarrow Q$ that uses the fact that P is false is called vacuous proof.

TRIVIAL PROOF:

A proof of $P \rightarrow Q$ that uses the fact that Q is true is called Trivial Proof.

We can quickly prove a Conditional Statement $P \rightarrow Q$ if we know that Conclusion is true (what the values of hypothesis).

Trivial Proofs are often important when special cases of theorems are proved.

Example: (Trivial Proof)

Let $P(n)$: If a and b are positive integers with $a \geq b$, then $a^n \geq b^n$

where domain: all integers
Show that $P(0)$ is true.

Sol:

$P(n)$: If $a \geq b$, then $a^n \geq b^n$: a & b are +ve integers

$P(0)$: If $a \geq b$, then $a^0 \geq b^0$

Here Conclusion $a^0 \geq b^0$ is true for +ve integers a and b .

~~So $P \rightarrow$~~

So, $P(0)$ is true, no matter what the ^{truth} value of hypothesis $a \geq b$.
This is an example of trivial proof.

Proof BY CONTRADICTION :

In a proof of $P \rightarrow Q$ by Contradiction, we assume that $\neg Q$ is true and then show that $\neg P$ must also be true (which was initially false) to arrive at Contradiction.

$P \rightarrow Q$ is true by Contradiction

First,

We assume that $\neg Q$ (negation of Conclusion) and Premise(s) P be true. and then arrive ~~to the~~ at the Contradiction that $\neg P$ to be true.

(P true and finally we have $\neg P$ true i.e. P is false : CONTRADICTION)

Proof by Contradiction Can be used to prove Conditional Statement ($P \rightarrow Q$).

Example #1 (Proof By Contradiction)

If $3n+2$ is odd, then n is odd

Sol: $P(n): 3n+2$ is odd
 $Q(n): n$ is odd

Start with the assumption:
 $\neg Q$ is true, (P is true)

i.e.

$\neg Q: n$ is not odd
: n is even

As n is even

$\Rightarrow n = 2K$

As we have $P(n): 3n+2$ is odd is true, But by taking n is even

$$3n+2 = 3(2K)+2$$

$$= 6K+2$$

$$= 2(3K+1)$$

$$= 2m \quad \text{which shows } P(n) \text{ is even i.e. } (\neg P \text{ is true})$$

$P(n)$ is false

Here we have P is true (Initial assumption) and $\neg P$ is also true.

Here we have a Contradiction to prove $P \rightarrow Q$, we suppose $\neg Q$ true is not correct. Q is true.

Example #2 (Proof By Contradiction)

Prove that $\sqrt{2}$ is irrational by giving proof by Contradiction.

Sol:-

Let $P : (\sqrt{2} \text{ is irrational})$

We suppose $\neg P$ is true i.e

$\sqrt{2}$ is not irrational i.e

$\sqrt{2}$ is rational

Then, there exist a and b with $\sqrt{2} = \frac{a}{b}$ ($a \nmid b$ with no common factor)
i.e $\frac{a}{b}$ in lowest terms

$$\sqrt{2} = \frac{a}{b}$$

Squaring both sides, we have

$$2 = \frac{a^2}{b^2}$$

$$\text{or } 2b^2 = a^2$$

$$2K = a^2 \quad \text{where } K = b^2 \text{ is some integer}$$

$$\Rightarrow a^2 \text{ is even } (a^2 = 2K)$$

Also a is even (we can use the fact that from definition of even integer)

$$\text{As } 2b^2 = a^2 \quad \text{and } a \text{ is even (i.e } a = 2m)$$

$$= (2m)^2$$

$$2b^2 = 4m^2$$

$$b^2 = 2\underline{m^2}$$

$$b^2 = 2K \approx \text{even and also } b \text{ is even}$$

As a and b are even (i.e 2 divides both a and b), which is contradiction that supposition of $\sqrt{2}$ is rational is wrong.

i.e $\sqrt{2}$ is irrational (proved).

$\sqrt{2} = \frac{a}{b}$ our supposition
with no common factor
if $a \nmid b$.
 a & b are in lowest
terms.

Proof of Equivalence

To prove a theorem that biconditional Statement ($P \leftrightarrow Q$), we show that both $P \rightarrow Q$ and $Q \rightarrow P$ are true.

Example #1:

Prove the theorem

"If n is positive integer, then
 n is odd if and only if n^2 is odd

Sol:-

we have to show that

If n is odd, then n^2 is odd, and

If n^2 is odd, then n is odd

$$\text{i.e. } (P \rightarrow Q) \wedge (Q \rightarrow P) = P \leftrightarrow Q$$

1st :

If n is odd, then n^2 is odd (Already shown in example 1 of Direct Proof Method)

2nd:

If n^2 is odd, then n is odd (Already shown in example #3 of Contrapositive Proof)

Because $P \rightarrow Q$ and $Q \rightarrow P$ are proved, So the theorem is true.

NOTE: In case several propositions to be equivalent.

To prove Propositions $P_1, P_2, \dots, P_n$ are equivalent, we have to show that

$$(P_1 \rightarrow P_2) \wedge (P_2 \rightarrow P_3) \wedge \dots \wedge (P_n \rightarrow P_1)$$

Example #2 (Equivalence Proof i.e. iff case)

Show that the following statements about integer n are equivalent.

P_1 : n is even

P_2 : $n-1$ is ~~is~~ odd

P_3 : n^2 is even

Sol:

We have to show that

$$(P_1 \rightarrow P_2) \wedge (P_2 \rightarrow P_3) \wedge (P_3 \rightarrow P_1) \leftrightarrow (P_1 \leftrightarrow P_2 \leftrightarrow P_3)$$

$P_1 \rightarrow P_2$: (Direct Method)

n is even

$$n = 2K$$

we have

$$\begin{aligned} n-1 &= 2K-1 \\ &= 2K-1+1-1 \\ &= 2K-2+1 = 2(K-1)+1 = 2m+1 \approx \text{odd} : \text{True} \end{aligned}$$

$P_2 \rightarrow P_3$: $(n-1)$ is odd, then n^2 is even

$n-1$ is odd

$$\text{so, } n-1 = 2K+1$$

$$\text{or } n = 2K+2 = \cancel{2(K+1)}$$

we have to show n^2

$$\begin{aligned} n^2 &= (2K+2)^2 = 4K^2 + 8K + 4 \\ &= 2(2K^2 + 4K + 2) \\ &= 2m \approx \text{even} : \text{True} \end{aligned}$$

$P_3 \rightarrow P_1$: n^2 is even, then $\overset{n \text{ is}}{\cancel{n-1}}$ even
(Contrapositive Proof)

$$\neg Q \rightarrow \neg P$$

~~$n-1$ is odd~~

n is odd

$$n = 2K+1$$

we have to show n^2 ,

$$\begin{aligned} n^2 &= (2K+1)^2 = 4K^2 + 4K + 1 \\ &= 2(2K^2 + 2K) + 1 = 2m+1 \approx \text{odd i.e. } \neg P \text{ is true} \end{aligned}$$

So $P_3 \rightarrow P_1$ is true

From all of above, we have proved $(P_1 \rightarrow P_2) \wedge (P_2 \rightarrow P_3) \wedge (P_3 \rightarrow P_1)$ is true
All are equivalent.

Counter Example :-

Show that

"Every +ve Integer is the sum of square of two integers" is false.

Sol:-

The above Statement is of the form :

$$\forall x P(x)$$

We need only to find a Counter example i.e x for which $P(x)$ is false.

We look for a Counter example, which is particular integer that is not the sum of square of two integers.

$$\text{Take } x = 3$$

3 Cannot be written as the Sum of Squares of two integers.

Proved it The Statement is false.

Mistakes In Proofs :

Visit page number 89